

MARCO STEFANELLI

Machine Learning Scientist | Earth Observation, Weather & Climate | Python, PyTorch & ML-Ready Data Pipelines

- Nationality: Italian
- Email: stfmrc.dev@gmail.com
- Portfolio: <https://stfmrc.github.io/MarcoStefanelli/>
- Languages: Italian (native), English (fluent), Croatian (beginner)

PROFESSIONAL SUMMARY

Machine-learning scientist with a background in physics and a PhD in Earth System and Climate Science, awarded a Marie Skłodowska-Curie COFUND postdoctoral fellowship ([SMASH](#)) to develop a neural-network-based observation operator for weather-radar data assimilation. Experienced in large-scale preprocessing of heterogeneous radar and numerical-model data; ML model design, training, and evaluation; from-scratch implementation of a 3DVar data-assimilation framework; integration of the learned observation operator into the 3DVar framework; and publication of reusable open-source software, pretrained model, and large-scale datasets. Able to lead projects from problem formulation and data preparation through experimentation, validation, implementation, and publication, while collaborating effectively across multidisciplinary scientific and operational teams.

TECHNICAL SKILLS

Programming and Machine learning: PyTorch, TensorFlow/Keras, deep learning, model implementation, training and evaluation, loss function design

Programming and data processing: Python, NumPy, xarray, GeoPandas, Rasterio, Bash, Fortran

Large-scale computing: GPU-accelerated workflows, CUDA, Linux/Unix, HPC, SLURM, PBS, LSF

Scientific data: TB-scale datasets of weather-radar, satellite, in-situ, and numerical-model data, primarily in NetCDF, GRIB, and HDF5 formats

Research engineering: Git, GitHub, Jupyter, reproducible preprocessing pipelines, dataset publication and documentation, code review, dependency and environment management

Testing and validation: Scientific validation scripts, consistency checks, diagnostic metrics, physical-plausibility assessment, and visualisation-based quality control

PROFESSIONAL EXPERIENCE

May 2026 – July 2026

Scientific Author (Contract), Faculty of Mathematics and Physics, University of Ljubljana

- Led and co-authored, with scientists at the Slovenian Environment Agency, the preparation of a manuscript documenting a five-year (2019–2023) raw-data collection of weather-radar reflectivity observations and ALADIN-SI model fields, released as five annual datasets and used to produce a separate curated ML-ready dataset; the manuscript is currently under review.

April 2024 – April 2026

Postdoctoral Researcher, SMASH (Marie Skłodowska-Curie COFUND), Faculty of Mathematics and Physics, University of Ljubljana

- Led the end-to-end research workflow, from problem formulation and data preparation through model development, experimentation, validation, and publication.
- Developed reproducible pipelines that aligned and transformed a five-year raw-data collection of weather-radar reflectivity observations and ALADIN-SI model fields into a curated, TB-scale ML-ready dataset for neural-network development and evaluation.
- Designed, trained, and validated a neural-network-based observation operator model in Python (PyTorch) using GPU-accelerated, CUDA-enabled workflows for assimilating weather-radar observations.
- Designed and implemented a 3DVar data-assimilation framework from scratch in Python and integrated the learned observation operator.

- Collaborated with researchers at the Slovenian Environment Agency to refine the integration of the neural-network observation operator within the 3DVar framework.
- Evaluated how alternative preprocessing and data-transformation strategies affected neural-network accuracy and subsequent data-assimilation performance.
- Collaborated with scientists from complementary disciplines at SISSA to address methodological and technical challenges in implementing the neural-network observation operator.
- Released open-source software and openly available datasets to support reproducibility and further scientific use.
- Developed diagnostic and validation scripts combining quantitative metrics with targeted visual analysis to assess model accuracy, physical consistency, and downstream data-assimilation performance.
- Led and co-authored the preprint “A Neural-Network Model-Measurement-Based Observation Operator For Weather Radar Reflectivity Assimilation” (manuscript currently under review).

February 2023 – March 2024

Researcher, CMCC Foundation – Euro-Mediterranean Center on Climate Change, Italy

- Processed and analysed large (TB-scale), heterogeneous observational datasets, including Argo profiling floats and satellite observations.
- Built reproducible Python and HPC workflows for large scientific datasets and operational modelling systems.
- Led and co-authored the publication “Data Assimilation for Advanced Cross-Scale Unstructured Grid Ocean Modelling.”

April 2018 – April 2019

Researcher, The Abdus Salam International Centre for Theoretical Physics (ICTP), Italy

- Designed computational experiments to study the effect of urbanisation on the West African monsoon using a regional climate model.
- Processed and analysed large regional climate-model and observational datasets, including precipitation and satellite measurements.
- Built Python/Linux workflows to compare, validate, and interpret model outputs against observations.

EDUCATION

November 2019 – January 2023

PhD, Future Earth, Climate Change and Societal Challenges — University of Bologna & CMCC, Italy

Dissertation: [Data assimilation for advanced cross-scale unstructured-grid ocean modelling](#).

- Introduced and implemented into the existing CMCC ocean 3DVar framework (OceanVar) a novel first-order recursive filter algorithm to model the horizontal component of the background error covariance matrix for unstructured grids.
- Used Bash and Fortran to couple the modified OceanVar 3DVar system with a regional implementation of the SHYFEM unstructured-grid ocean model.
- Processed and analysed Argo profiling float and satellite altimetry data for use in the established ocean forecasting system.
- Designed validation workflows comparing numerical-model outputs against observational data.
- Collaborated with the Data Science Group at Imperial College London on methodological aspects of neural-network development and evaluation.

September 2014 – March 2018

MSc, Earth System Physics — University of Trieste, Italy

Dissertation: A study of urbanisation effects on the West African monsoon.

SELECTED PUBLICATIONS AND PREPRINTS

- Stefanelli, M., Jansen, E., Aydoğdu, A., Federico, I., Coppini, G. (2025). Frontiers in Marine Science. [Data Assimilation for Advanced Cross-Scale Unstructured Grid Ocean Modelling](#).
- Preprint (arXiv, under review at GMD): Stefanelli, M., Zaplotnik, Ž., Skok, G. (2026) [A Neural-Network Model-Measurement-Based Observation Operator For Weather Radar Reflectivity Assimilation](#).

OPEN-SOURCE SOFTWARE

- [Radar-ALADIN data preprocessing](#) - Python/Bash
Reusable data preprocessing workflow for combining radar observations and numerical-model data for ML use.
- [3DVar neural network-based observation operator](#) - Python
Implementation of the NN observation operator developed during the Marie Skłodowska-Curie (SMASH) project. A pre-trained model is also available in the same repository.
- [3DVar integration of a neural network-based observation operator](#) - Python
Open-source, from-scratch implementation of a 3DVar framework integrating the learned observation operator for weather-radar assimilation.

PUBLISHED DATASETS

- [LISCA-ALADIN HNN](#) - Curated ML-ready dataset

Curated NetCDF dataset spanning 2019–2023, derived from aligned weather-radar reflectivity observations and ALADIN-SI model fields. Ready for neural-network development and evaluation. Available on Zenodo with a persistent DOI.

- [2019](#) | [2020](#) | [2021](#) | [2022](#) | [2023](#) - Raw-data collection

Five annual dataset releases containing the source weather-radar reflectivity observations in HDF5 format and ALADIN-SI model fields in GRIB format. All are available on Zenodo with persistent DOIs.

SELECTED PRESENTATIONS

- European Geosciences Union (EGU), Vienna (2022 and 2025)
2022: [Variational data assimilation for advanced cross-scale ocean modelling](#).
2025: [A neural network-based observation operator for weather radar data assimilation](#)
- European Meteorological Society (EMS) Annual Meeting, Ljubljana (2025)
Video: [A neural network-based observation operator for weather radar data assimilation](#)
- Slovenian Meteorology Society, Ljubljana (2025)
Video: [A neural network-based observation operator for weather radar data assimilation](#)

ADDITIONAL INFORMATION

Science communication: Co-developed educational activities with secondary-school teachers and delivered educational activities on climate change, scientific literacy, and misinformation for secondary-school students. Created an interactive [JavaScript tool](#) based on the Lorenz-63 model to demonstrate deterministic chaos and how data assimilation reduces forecast uncertainty.